The Spark: Artificial Neurons Fire Up AI

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University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - Deep learning applications
  - History of neural networks and deep learning
  - How does a machine learn?
  - Course logistics
- Assignments (Canvas)
  - Lab assignment 1 due in 1.5 weeks
- Questions?

Today's Topics

- Biological inspiration (late 1800s & early 1900s): discovery of neurons
- Biological modeling (1957): mimicking "intelligence" to understand it
- Crisis (1969): models proven incapable of many forms of "intelligence"
- Programming tutorial: basics of PyTorch

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Motivation: How Do We Sense and Respond to the World?

Who has experienced something that is:

"hot"



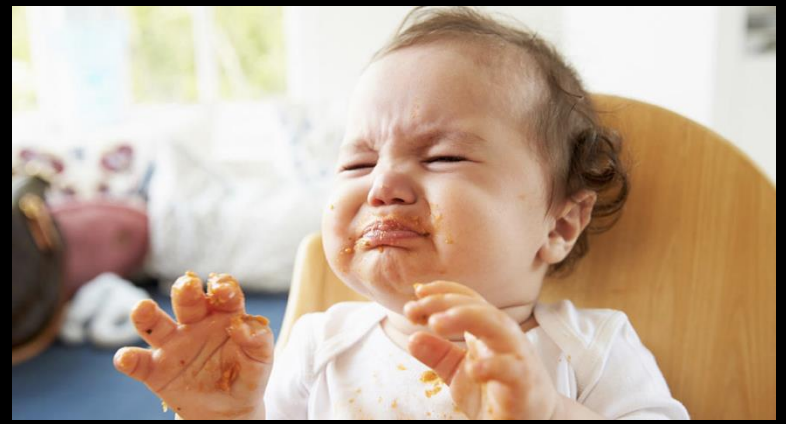
https://www.clipart.email/clipart/dont-touch-hot-stove-clipart-73647.html

"loud"



https://kisselpaso.com/if-the-sun-city-music-fest-gets-too-loud-there-is-a-phone-number-you-can-call-to-complain/

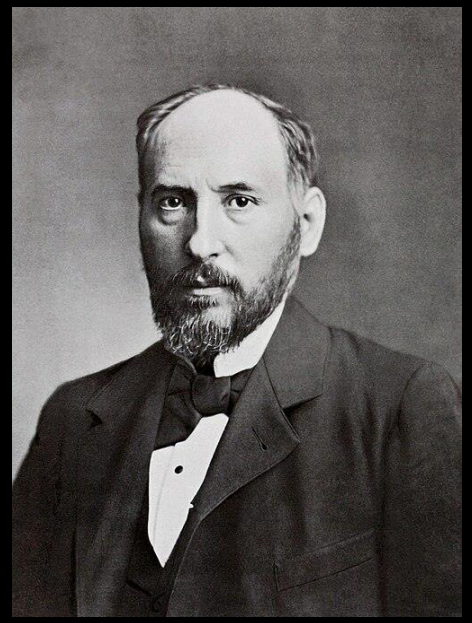
"spicy"



https://www.babycenter.com/404_when-can-my-baby-eat-spicy-foods_1368539.bc

What is that enables us to have such experiences?

Discovery: Neurons Are the Basic Unit for Sensing and Responding to the World



Santiago Ramón y Cajal
1852-1934
"Father of modern neuroscience"

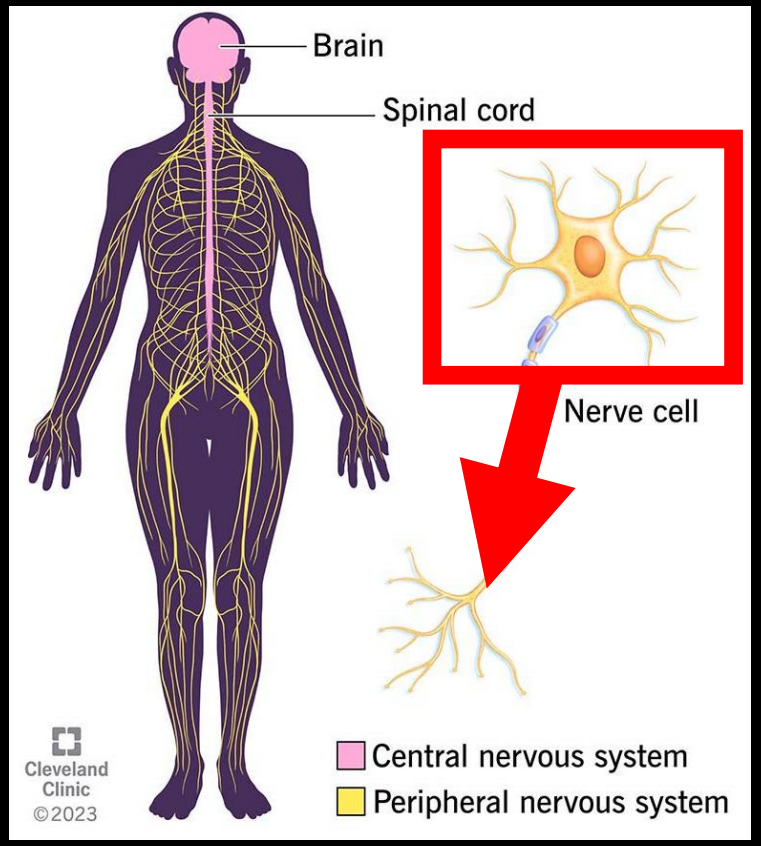
Awarded 1906 Nobel Prize for discovering in the late 1880s organisms' nervous systems consist of independent cells (called neurons) that communicate across small gaps



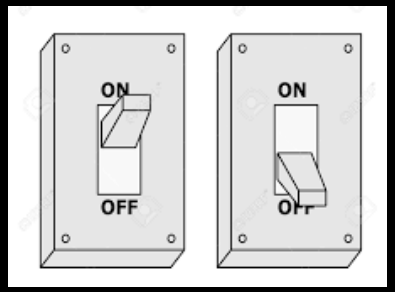
Edgar Adrian
1889-1977
A "father" of modern neurophysiology

Awarded 1932 Nobel Prize for using his new recording device to reveal how neurons communicate, including the all-or-none firing behavior of neurons

Discovery: Neurons Are the Basic Unit for Sensing and Responding to the World



If incoming signals exceed a certain threshold within a short period of time, the neuron “fires”

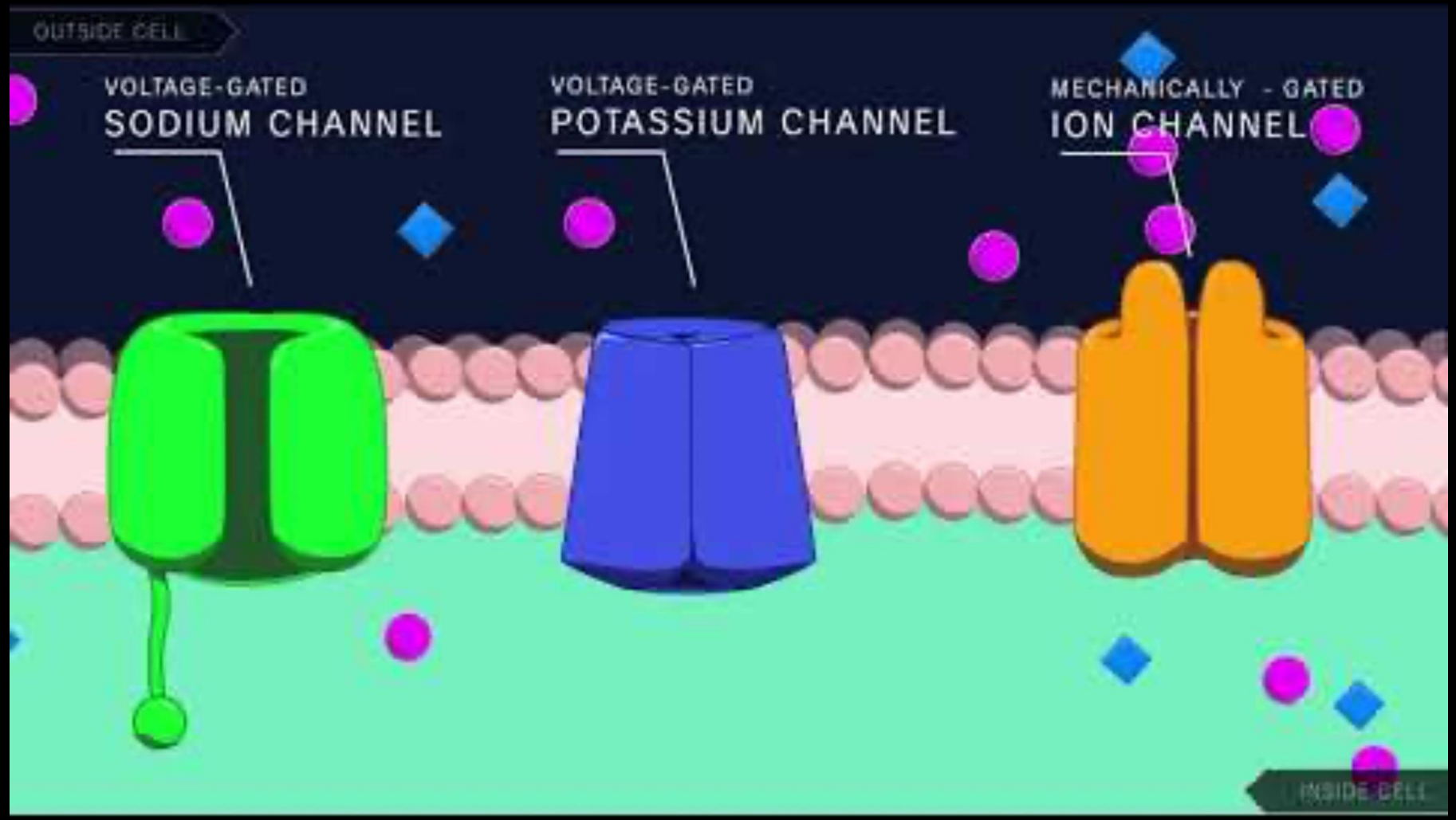


Analogous to turning on a light switch!

Awarded 1906 Nobel Prize for discovering in the late 1880s organisms’ nervous systems consist of independent cells (called neurons) that communicate across small gaps

Awarded 1932 Nobel Prize for using his new recording device to reveal how neurons communicate, including the all-or-none firing behavior of neurons

Basics of Biological Neurons



0:08-1:12 minutes of https://youtu.be/oa6rvUJlg7o?si=HIVHkOMvHEWo7Mr2&t=8

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Biological Modeling



Frank Rosenblatt
1928-1971
Psychologist

Report No. 85-460-1

THE PERCEPTRON

A PERCEIVING AND RECOGNIZING AUTOMATON

(PROJECT PARA)

January, 1957

"new research program... objective of designing, fabricating, and evaluating an electronic brain model... capable of "learning" responses... capable of concept formation, language translation, collation of military intelligence, and the solution of problems through inductive logic."

Context: Availability of Architecture, Hardware, and Learning *Concepts*

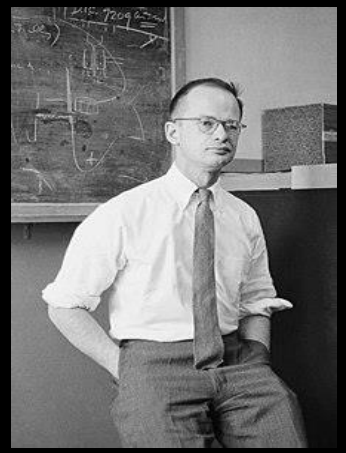
- 1943: model of biological neuron invented
- 1945: first digital computer created
- 1949: learning paradigm where "neurons that fire together, wire together" introduced in Hebb's "The Organization of Behavior"

Architecture Background: Neuron Model

Warren McCulloch
1898-1969
Neurophysiologist



Walter Pitts
1923-1969
Mathematician



- All-or-none neuron firing = propositional logic
  - The study of relationships between statements (propositions) to decide if they are true or false
- Truth tables exemplify how statements combine to create logical operations; e.g., what are these?

P	Q	
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	
T	T	T
T	F	T
F	T	T
F	F	F

And many more here: https://sites.millersville.edu/bikenaga/math-proof/truth-tables/truth-tables.html

Warren McCulloch and Walter Pitts, A Logical Calculus of Ideas Immanent in Nervous Activity, 1943

Architecture Background: Key Ideas

"… our knowledge of the world, including ourselves, is incomplete as to space and indefinite as to time. This ignorance, implicit in all our brains, is the counterpart of the abstraction which renders our knowledge useful. The role of brains in determining the epistemic relations of our theories to our observations and of these to the facts is all too clear, for it is apparent that every idea and every sensation is realized by activity within that net… There is no theory we may hold and no observation we can make that will retain so much as its old defective reference to the facts if the net be altered… With determination of the net, the unknowable object of knowledge, the "thing in itself," ceases to be unknowable."

Do you agree with the authors' following conclusions?

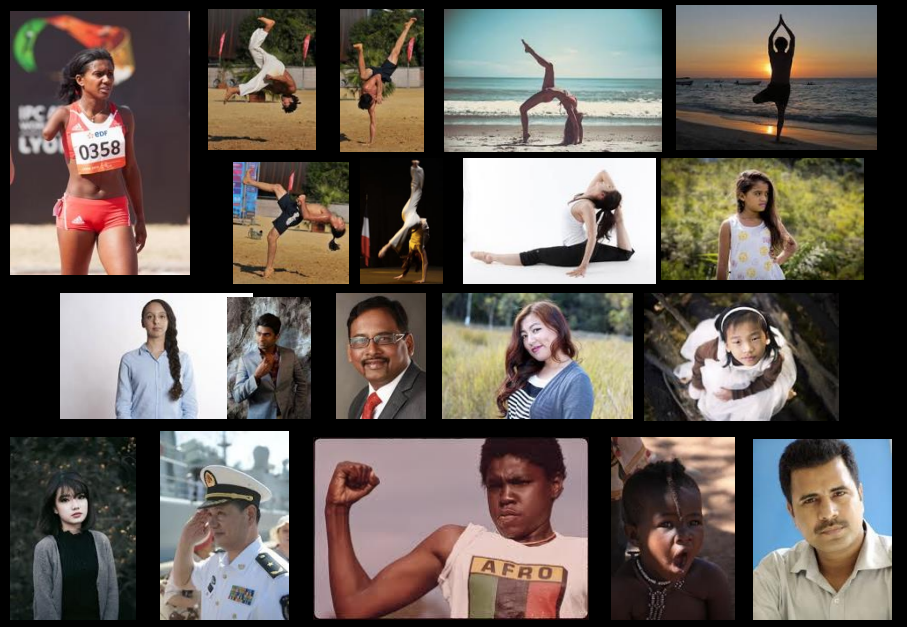
1. Brains don't sense everything
2. Our brains, instead, compress and abstract information into useful patterns (e.g., for deciding how to act)
3. Thinking and experiences are just neural computation between many simple computational devices
4. Specification of a brain's computational network dictates reality (e.g., brain damage changes facts)

Warren McCulloch and Walter Pitts, A Logical Calculus of Ideas Immanent in Nervous Activity, 1943

Architecture: Key Ideas

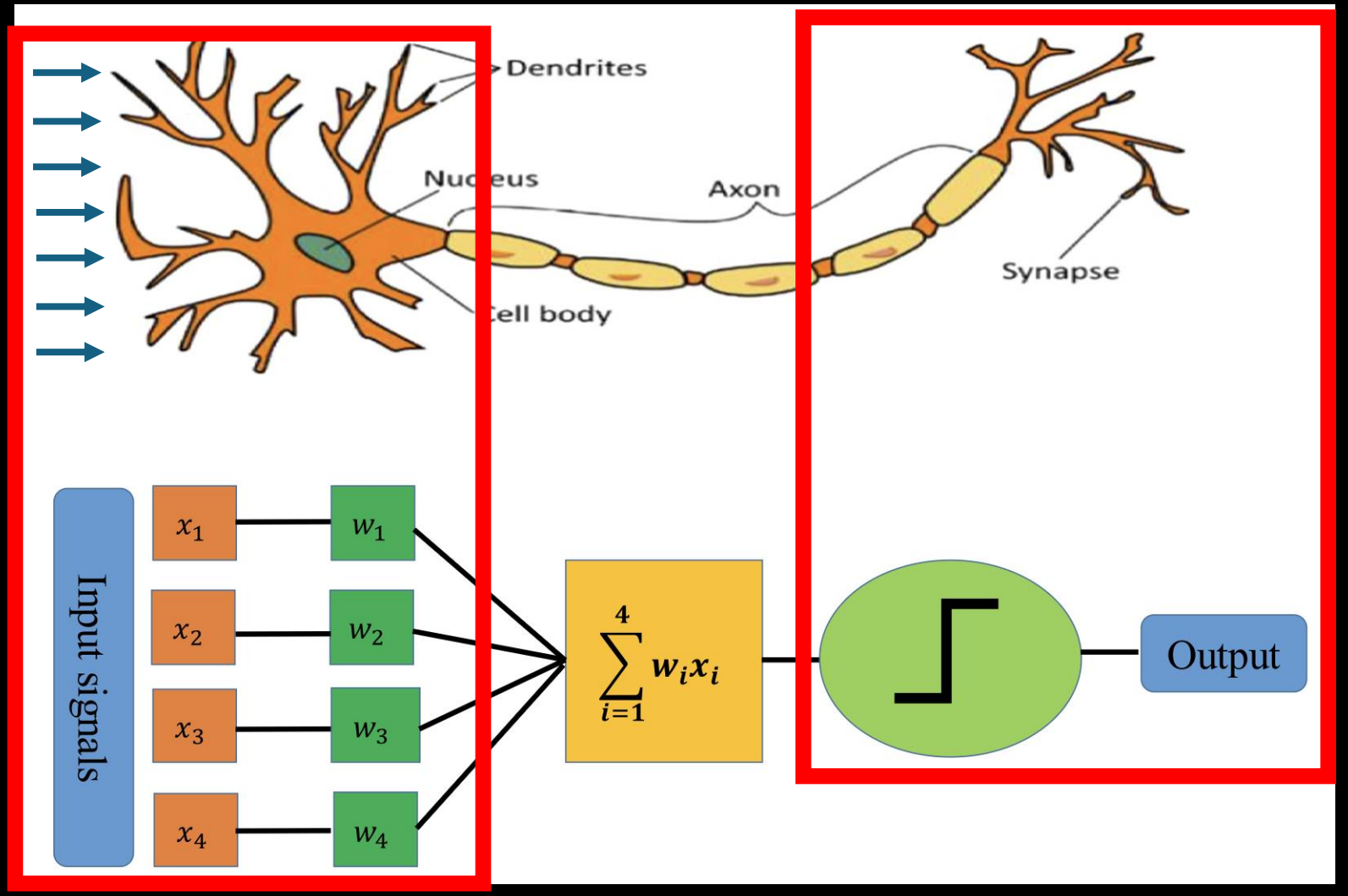
- It is not feasible to perform brute-force search, comparing a new example against everything else in the world for a match, since computers' memory is finite





- Learning is necessary in order to generate compact, generalizable patterns that can fit into finite memory (just like humans)

Architecture: Mimics Biological Neuron



Extends McCulloch-Pitts neuron by letting inputs and weights be any value and modifiable

(“fires” only if aggregate value exceeds threshold)

[Ghasemian, Razminia, and Rostami. Quantum Information Processing 2023]

Architecture: Mathematical Representation

- Function deciding output value ("fire" or not):

$$\phi(z) = \begin{cases} 1 & if\ z \geq \theta \\ -1 & otherwise \end{cases}$$

- Rewriting function:

$$\phi(z) = \begin{cases} 1 & if\ z \geq 0 \\ -1 & otherwise \end{cases}$$

- Where:

$$z = w_0 x_0 + w_1 x_1 + \ldots + w_m x_m = \boldsymbol{w}^T \boldsymbol{x}$$

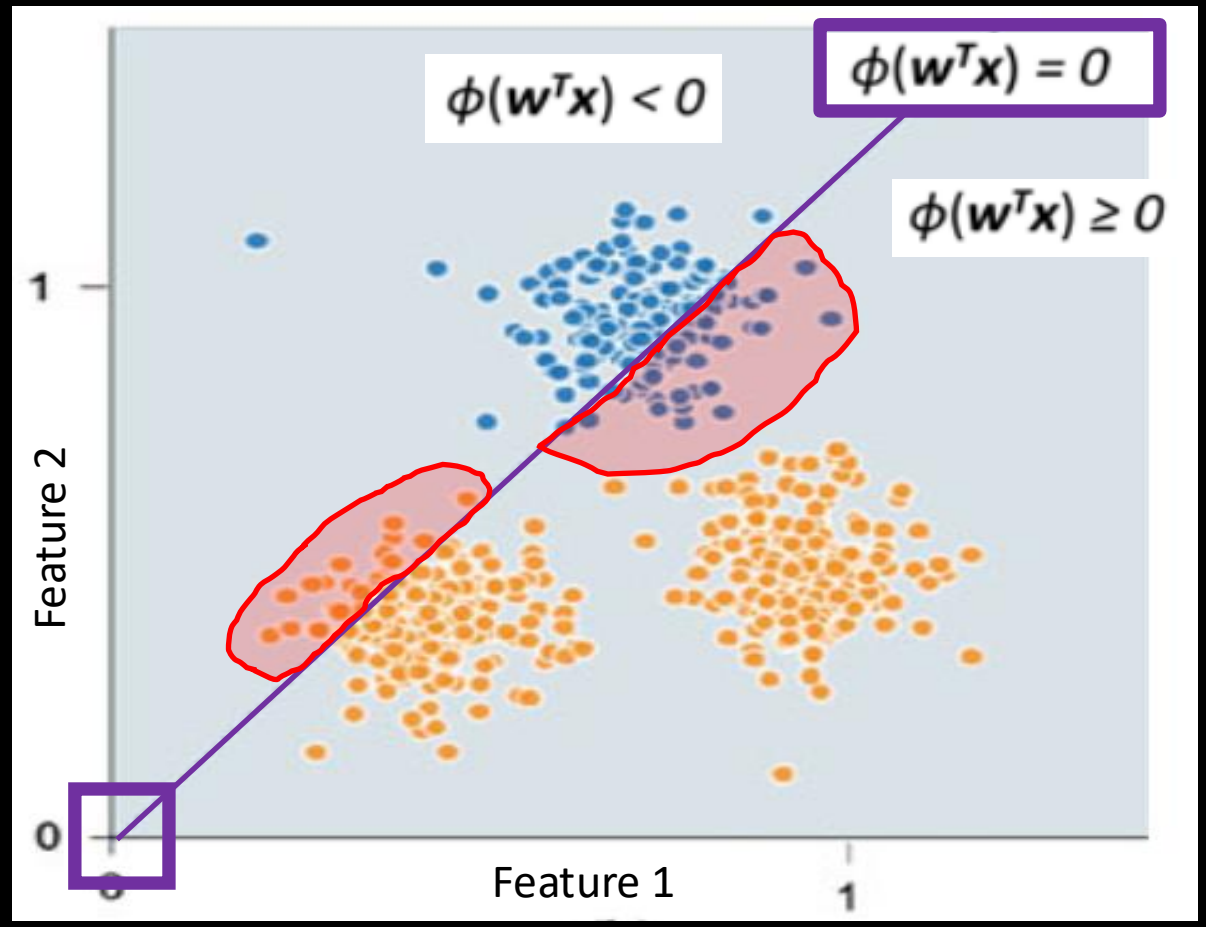
Bias $-\theta$ 1

(indicates how easy/hard it is to get the perceptron to "fire"; this value is learned)

Note: output values of {-1, 1} and {0, 1} are both used, including in Ch 2.5 and 4 of the Kamath book. These are paired with thresholds of 0 and 0.5 respectively.

Architecture: Simple Graphical Example

Binary classification: decision boundary separates orange and blue dots



What errors does the model make?

Mathematically, the bias term enables the model to NOT pass through the origin

Kamath et al. Deep Learning for NLP and Speech Recognition. 2019.

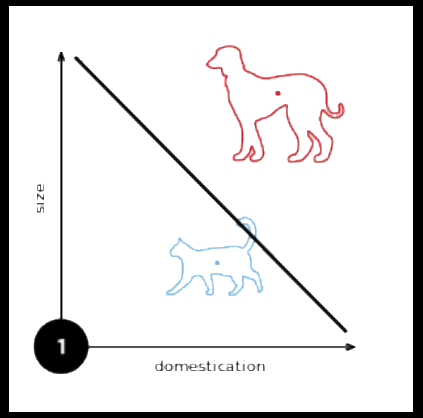
Learning Algorithm: Recall Motivation

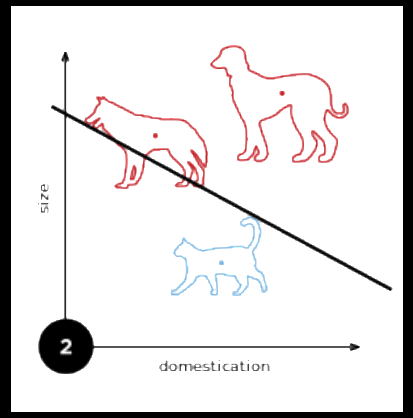
- Rather than storing all experiences, the algorithm abstracts useful patterns from them as changes to neural network parameters

Learning Algorithm: Mistake-Driven Rule

- Approach: for each example, the model makes a prediction. After an error, its weights are updated to try to place the misclassified example on the correct side.

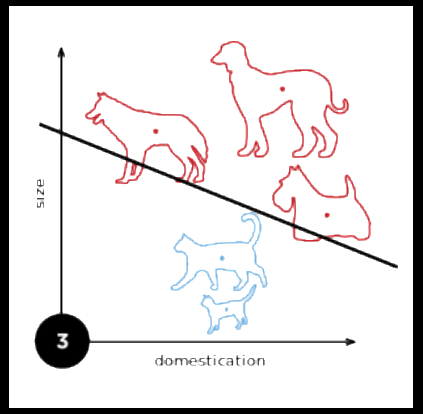
Assume this learned model

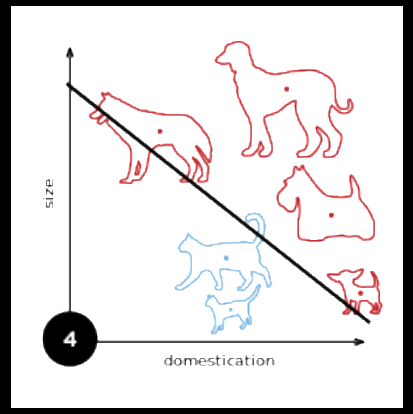




Would old model make a mistake?
- Yes; model changed

Would old model make a mistake?
- No; model didn't change





Would old model make a mistake?
- Yes; model changed

https://en.wikipedia.org/wiki/Perceptron

Learning Algorithm: Mistake-Driven Rule

- Approach: for each example, the model makes a prediction. After an error, its weights are updated to try to place the misclassified example on the correct side.

- Implementation, which can run on hardware (bridges theory with application):
  - Repeat until stopping criterion met:
    - For each input example, **x**, compute model's prediction: $\hat{y} = \begin{matrix} 1 & \text{if } \mathbf{w}^\top \mathbf{x} \geq 0 \\ -1 & \text{otherwise} \end{matrix}$
    - Update all model weights as follows: $w \leftarrow w + \eta \left(y_{true} - y_{pred} \right) x$

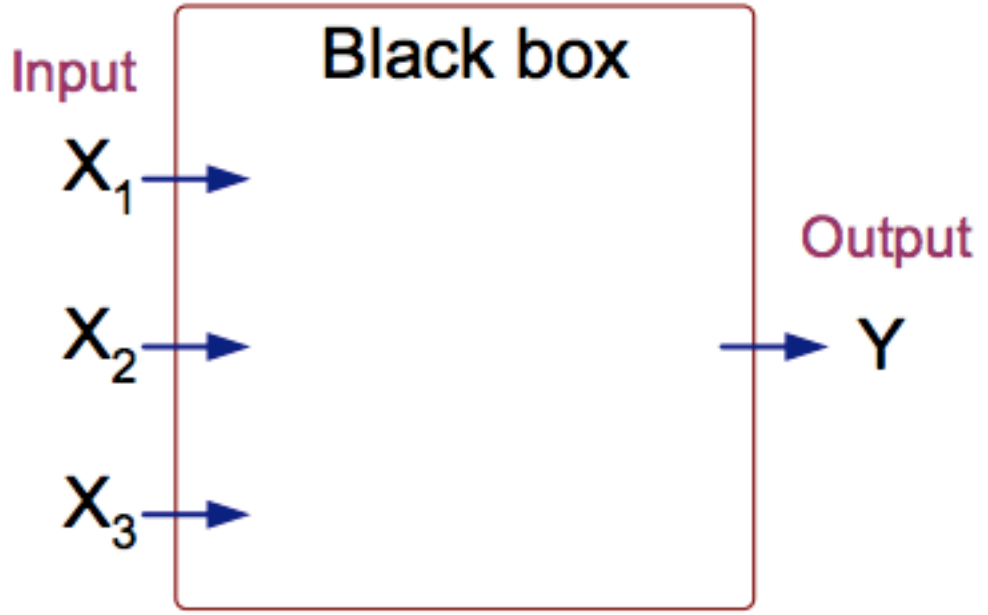
Learning Rate (set a priori)

- Convergence guarantees provided; correct solution will be found in some conditions

Example: True Model

- Y is 1 if at least 2 of the 3 inputs are 1, and -1 otherwise

X_1	X_2	X_3	Y
1	0	0	?
1	0	1	
1	1	0	
1	1	1	
0	0	1	
0	1	0	
0	1	1	
0	0	0	

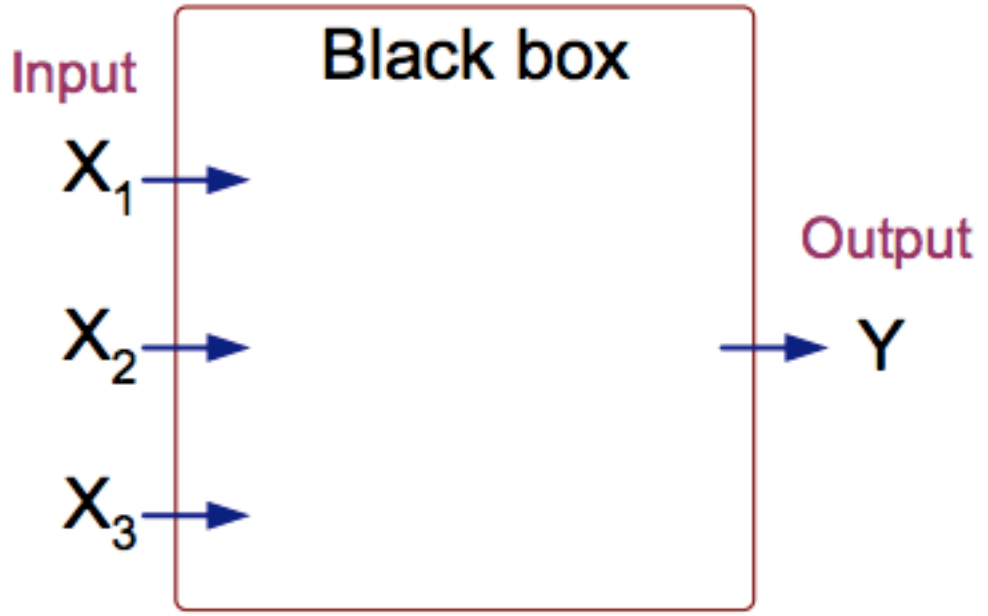


https://www-users.cs.umn.edu/~kumar001/dmbook/slides/chap4_ann.pdf

Example: True Model

- Y is 1 if at least 2 of the 3 inputs are 1, and -1 otherwise

X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	
1	1	0	
1	1	1	
0	0	1	?
0	1	0	
0	1	1	
0	0	0	

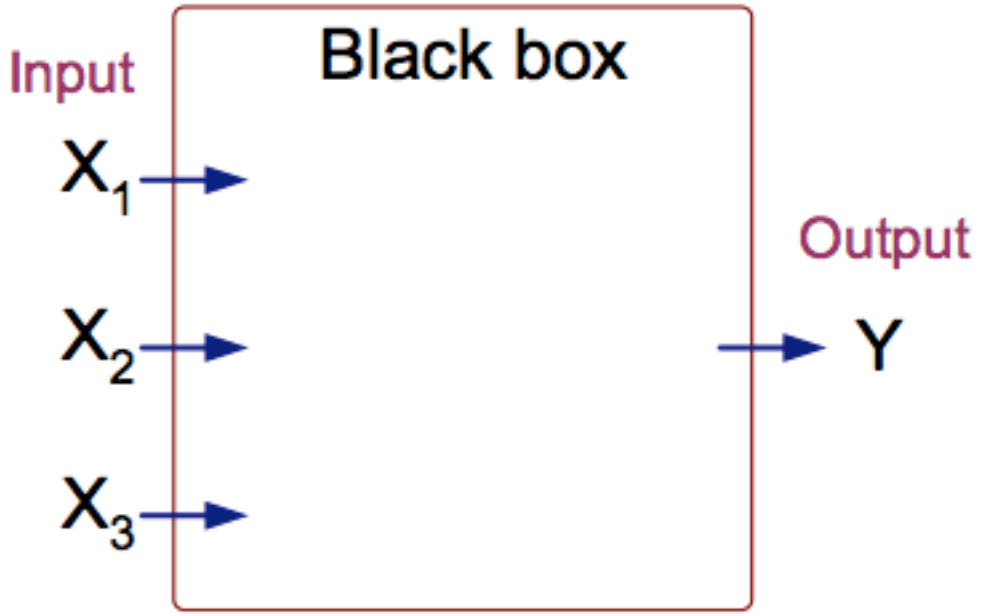


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Example: True Model

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X_1	X_2	X_3	Y
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1	0	1	1
1	1	0	?
1	1	1	
0	0	1	
0	1	0	
0	1	1	
0	0	0	

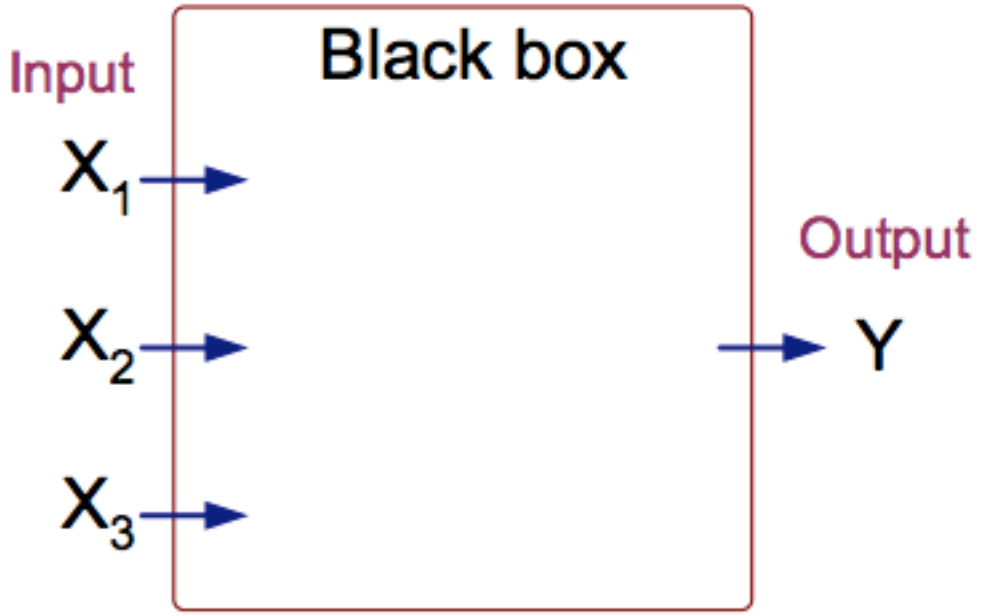


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Example: True Model

- Y is 1 if at least 2 of the 3 inputs are 1, and -1 otherwise

X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1



https://www-users.cs.umn.edu/~kumar001/dmbook/slides/chap4_ann.pdf

Example: Training with 1rst Sample

- Compute predicted value:

X_1	X_2	X_3	Y	Predicted
1	0	0	-1	?

w_0	w_1	w_2	w_3
0	0	0	0

$$\hat{y} = \begin{matrix} 1 & \text{if } \mathbf{w}^T\mathbf{x} \geq 0 \\ -1 & \text{otherwise} \end{matrix}$$

What does the model predict?

$$\mathbf{w}^T\mathbf{x} = 1 \text{ x } w_0 + \text{x}_1 \text{ x } w_1 + \text{x}_2 \text{ x } w_2 + \text{x}_3 \text{ x } w_3 \; = \; 1\text{x}0 \; + \; 1\text{x}0 \; + 0\text{x}0 + 0\text{x}0 \; = \; 0$$

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Example: Training with 1rst Sample

- Update parameters: $w \leftarrow w + \eta \left(y_{true} - y_{pred}\right) x$

X_1	X_2	X_3	Y	Predicted
1	0	0	-1	1

w_0	w_1	w_2	w_3
0	0	0	0

$$w_0 \leftarrow w_0 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right)$$

$$w_1 \leftarrow w_1 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right) x_1$$

$$w_2 \leftarrow w_2 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right) x_2$$

$$w_3 \leftarrow w_3 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right) x_3$$

Will the model parameters get updated?

https://www-users.cs.umn.edu/~kumar001/dmbook/slides/chap4_ann.pdf

Example: Training with 1rst Sample

- Update parameters: $w \leftarrow w + \eta \left(y_{true} - y_{pred} \right) x$

X_1	X_2	X_3	Y	Predicted
1	0	0	-1	1

w_0	w_1	w_2	w_3
0	0	0	0
?	?	?	?

Assume learning rate = 0.1:

$$w_0 = 0 + 0.1\,(-1 - 1) = -0.2$$

$$w_1 = 0 + 0.1\,(-1 - 1)\,\text{x}\,1 = -0.2$$

$$w_2 = 0 + 0.1\,(-1 - 1)\,\text{x}\,0 = 0$$

$$w_3 = 0 + 0.1\,(-1 - 1)\,\text{x}\,0 = 0$$

Example: Training with 2nd Sample

- Compute predicted value:

X_1	X_2	X_3	Y	Predicted
1	0	0	-1	1
1	0	1	1	?

w_0	w_1	w_2	w_3
0	0	0	0
-0.2	-0.2	0	0

$$\hat{y} = \begin{matrix} 1 & \text{if } \mathbf{w}^T\mathbf{x} \geq 0 \\ -1 & \text{otherwise} \end{matrix}$$

What does the model predict?

$$\mathbf{w}^T\mathbf{x} = 1 \text{ x } w_0 + \text{x}_1 \text{ x } w_1 + \text{x}_2 \text{ x } w_2 + \text{x}_3 \text{ x } w_3 = 1\text{x} - 0.2 + 1\text{x} - 0.2 + 0\text{x}0 + 1\text{x}0 = -0.4$$

Example: Training with 2nd Sample

- Update parameters: $w \leftarrow w + \eta \left(y_{true} - y_{pred}\right) x$

X_1	X_2	X_3	Y	Predicted
1	0	0	-1	1
1	0	1	1	-1

w_0	w_1	w_2	w_3
0	0	0	0
-0.2	-0.2	0	0

$$w_0 \leftarrow w_0 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right)$$

$$w_1 \leftarrow w_1 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right) x_1$$

$$w_2 \leftarrow w_2 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right) x_2$$

$$w_3 \leftarrow w_3 + \eta \left(y_{\text{true}} - y_{\text{pred}}\right) x_3$$

Will the model parameters get updated?

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Example: Training with 2nd Sample

- Update parameters: $w \leftarrow w + \eta \left(y_{true} - y_{pred}\right) x$

X_1	X_2	X_3	Y	Predicted
1	0	0	-1	1
1	0	1	1	-1

w_0	w_1	w_2	w_3
0	0	0	0
-0.2	-0.2	0	0
?	?	?	?

$$w_0 = -0.2 + 0.1\,(1 - -1) = 0$$

$$w_1 = -0.2 + 0.1\,(1 - -1)\,\text{x}\,1 = 0$$

$$w_2 = 0 + 0.1\,(1 - -1)\,\text{x}\,0 = 0$$

$$w_3 = 0 + 0.1\,(1 - -1)\,\text{x}\,1 = 0.2$$

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Example: Training for One Epoch

- For every training example:
  - Predict value
  - Update parameters
- How many examples are in this training dataset?
- Why might the weights not change for examples 2-4?

X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1

	w_0	w_1	w_2	w_3
0	0	0	0	0
1	-0.2	-0.2	0	0
2	0	0	0	0.2
3	0	0	0	0.2
4	0	0	0	0.2
5	-0.2	0	0	0
6	-0.2	0	0	0
7	0	0	0.2	0.2
8	-0.2	0	0.2	0.2

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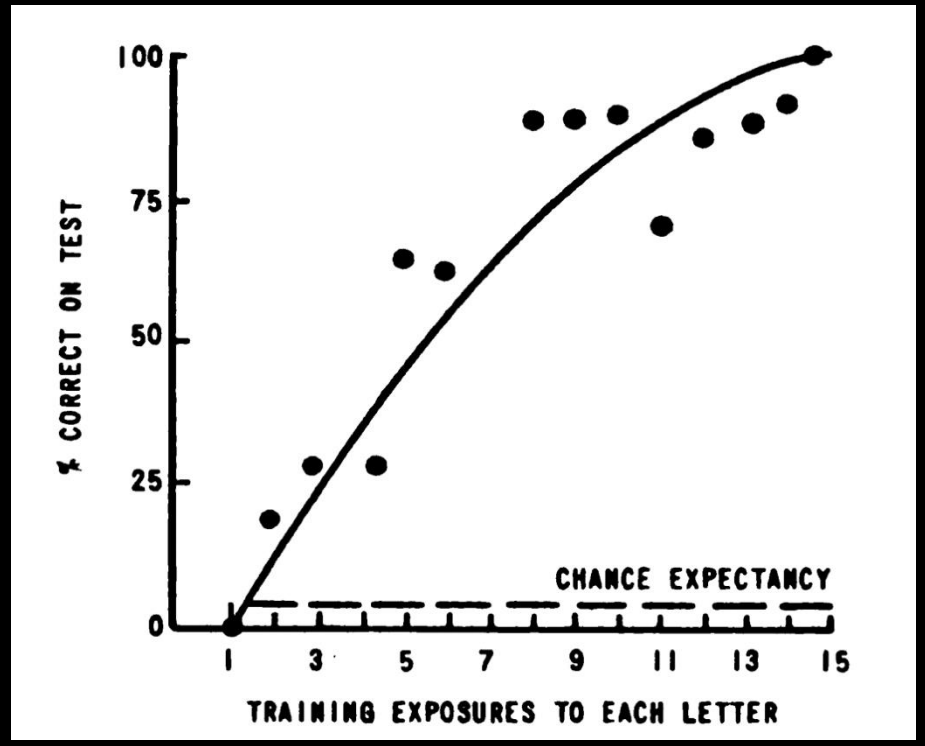
Example: Training for Multiple Epochs

- Continue repeatedly applying these two steps with every example

Epoch	w_0	w_1	w_2	w_3
0	0	0	0	0
1	-0.2	0	0.2	0.2
2	-0.2	0	0.4	0.2
3	-0.4	0	0.4	0.2
4	-0.4	0.2	0.4	0.4
5	-0.6	0.2	0.4	0.2
6	-0.6	0.4	0.4	0.2

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Experiments: Rosenblatt's Perceptron Learned!



e.g., Learning curve for 26 letters in the alphabet, presented in fixed positions

(and many more in Rosenblatt's 1962 book)



Mark I Perceptron built in 1960 for more experiments (e.g., camera sees "C")

(machine now at the Smithsonian museum)

https://en.wikipedia.org/wiki/Perceptron

Rosenblatt's Motivations

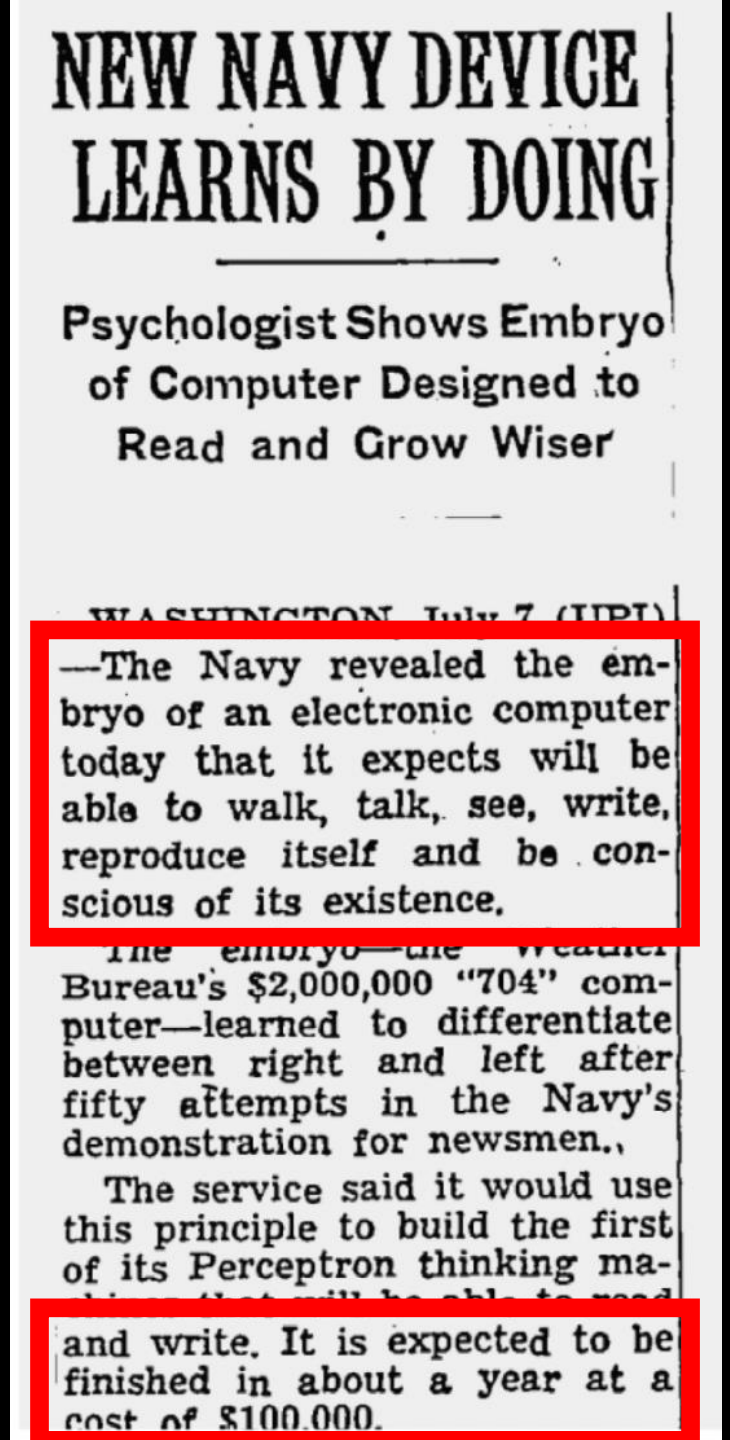
"For this writer, the perceptron program is not primarily concerned with the invention of devices for 'artificial intelligence', but rather with investigating the physical structures and neurodynamic principles which underlie 'natural intelligence'. A perceptron is first and foremost a brain model, not an invention for pattern recognition. As a brain model, its utility is in enabling us to determine the physical conditions for the emergence of various psychological properties. It is by no means a "complete" model, and we are fully aware of the simplifications which have been made from biological systems…"

[Rosenblatt. Spartan books 1962]

Others Had Other Aspirations; e.g.,

New York Times article, July 8, 1958 :

https://www.nytimes.com/1958/07/08/archives/new-navy-device-learns-by-doing-psychologist-shows-embryo-of.html



NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo of Computer Designed to Read and Grow Wiser

—The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

Bureau's $2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking ma-

and write. It is expected to be finished in about a year at a cost of $100,000.

Some Achieved Commercial Success…

e.g.,



Our optical reader can do anything your keypunch operators do.

(Well, almost.)

It can't make time on company time. Or use the office for intimate tete-a-tetes. Or be a social butterfly. But it *can* read. And gobble data at the rate of 2400 typewritten characters a second. And reduce errors from a keypunch operator's one in a thousand to an efficient one in a *hundred* thousand.

Our machine reads the same stuff a roomful of keypunch operators read, only better. And at a much lower cost. It has an Electronic Retina that works like the human eye. And it computes while it reads. It checks up on itself and on the material it's reading.

Right now, our Electronic Retina Computing Reader is saving people like United Air Lines hundreds of dollars every day. It could replace your entire keypunch department, which costs your company about $8,000 per keypunch operator per year.

Count your keypunch operators and ask your Financial VP what this would save your company. Then drop us a line.

RECOGNITION EQUIPMENT Incorporated

U.S. Headquarters: Dallas, Texas 214-823-8194 Offices in principal U.S. cities and Frankfurt, London, Milan, Paris and Stockholm

https://www.ebay.ca/itm/335521091499#

And Many Expressed Fear…

"In from three to eight years we will have a machine with the general intelligence of an average human being. I mean a machine that will be able to read Shakespeare, grease a car, play office politics, tell a joke, have a fight. At that point the machine will begin to educate itself with fantastic speed. In a few months it will be at genius level and a few months after that its powers will be incalculable… Once the computers got control, we might never get it back. We would survive at their sufferance. If we're lucky, they might decide to keep us as pets… I have warned [people in the Pentagon] again and again that we are getting into very dangerous country. They don't seem to understand."
– attributed to Marvin Minsky, controversially; "Father of AI"

"When I checked Minsky's prophecy with other people working on Artificial Intelligence, however, many of them said that Minsky's timetable might be somewhat wishful—"give us 15 years," was a common remark—but all agreed that there would be such a machine and that it could precipitate the third Industrial Revolution, wipe out war and poverty and roll up centuries of growth in science, education and the arts. At the same time a number of computer scientists fear that the godsend may become a Golem [an uncontrollable creation]."
- Journalist

1970 LIFE article



https://gwern.net/doc/reinforcement-learning/robot/1970-darrach.pdf

Today's Topics

- Biological inspiration (late 1800s & early 1900s): discovery of neurons
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- Crisis (1969): models proven incapable of many forms of "intelligence"
- Programming tutorial: basics of PyTorch

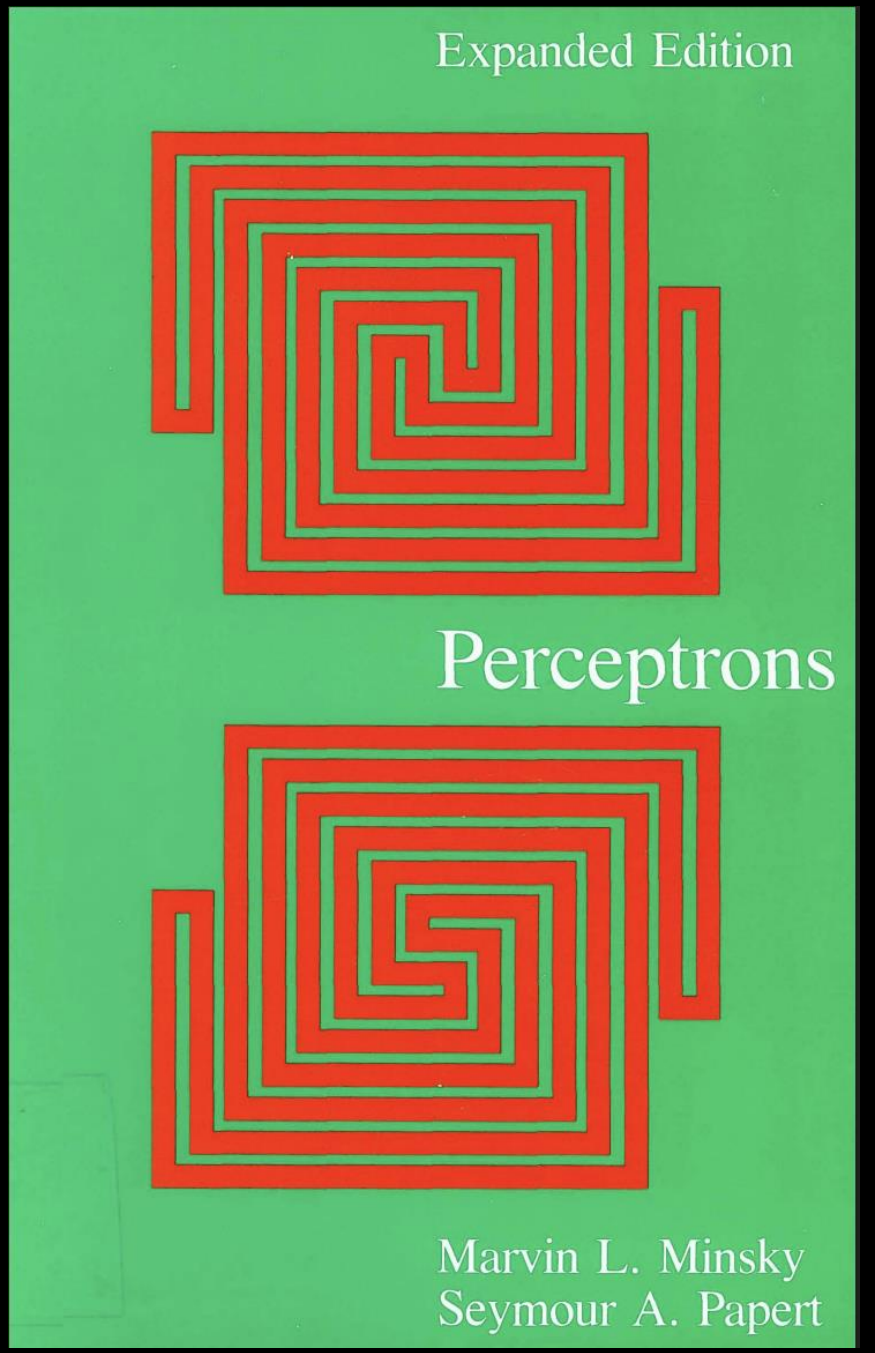
Main Instigator



https://www.wyso.org/2016-08-05/remembering-a-thinker-who-thought-about-thinking

Marvin Minsky
1927-2016
MIT Professor
1969 Turing Award for co-founding the AI field

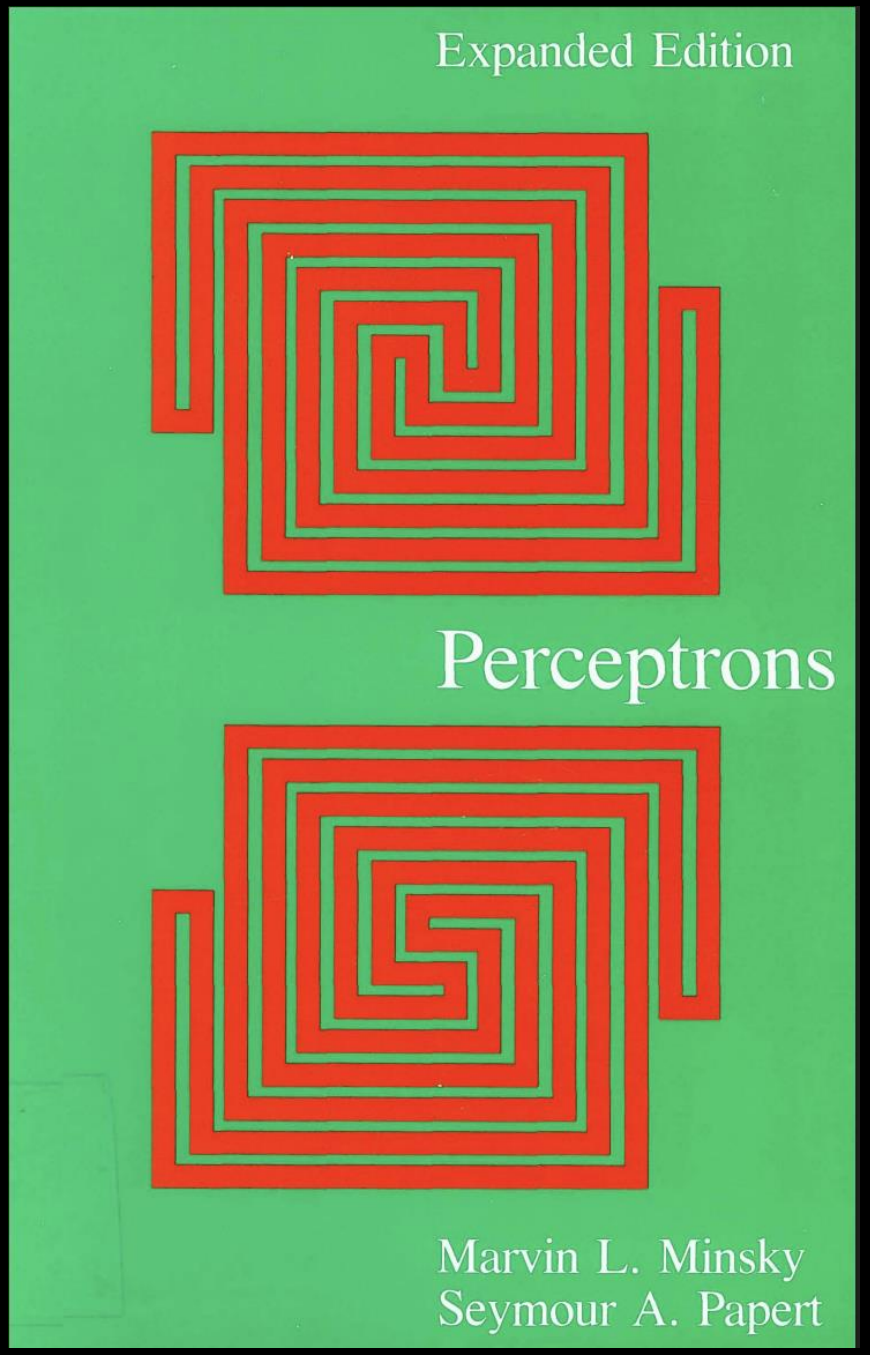
Seymore Papert
1928-2016
MIT Professor



Main Instigator

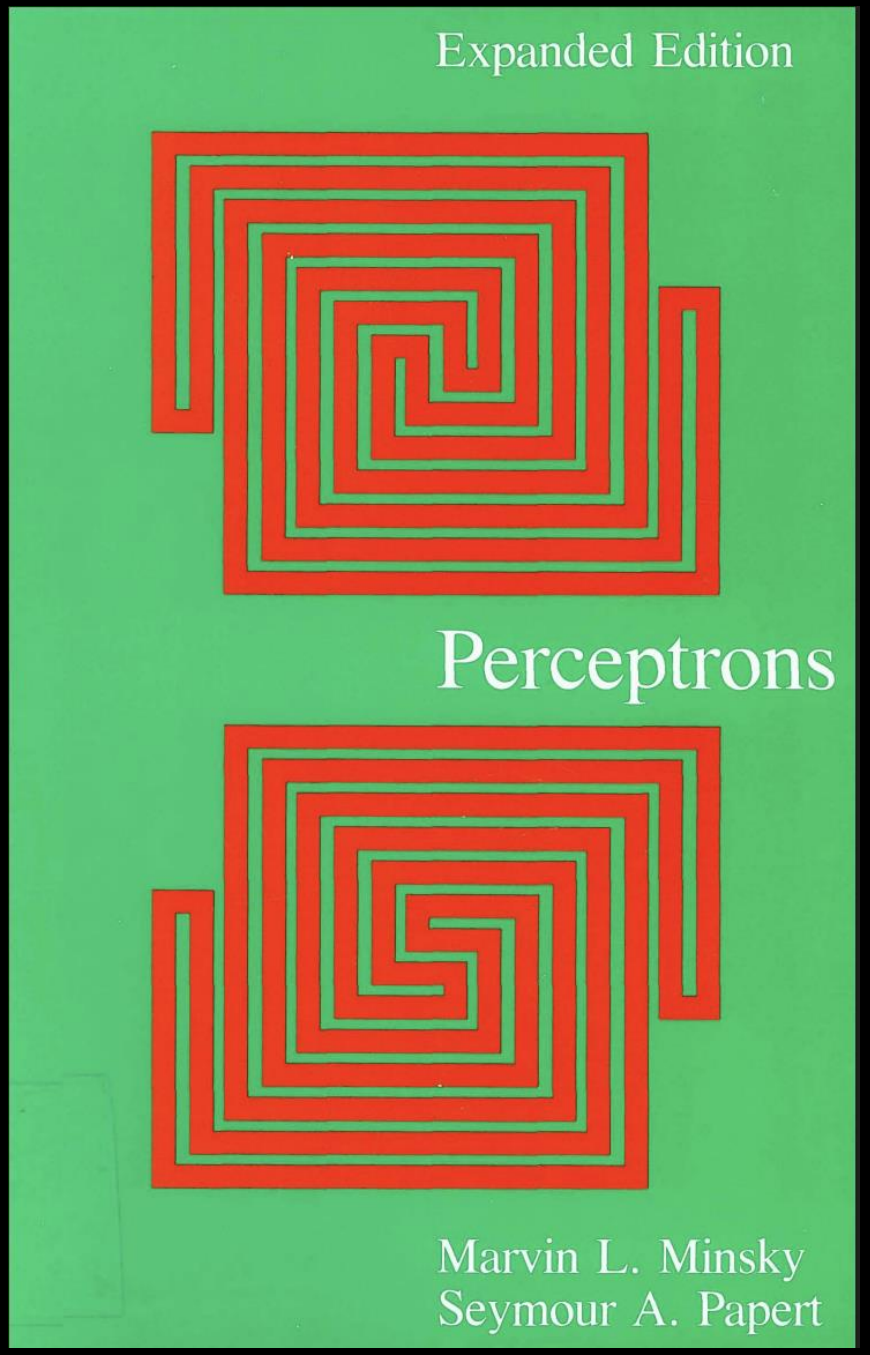
"Our collaboration began in 1963 when we were brought together by Warren S. McCulloch."

"… first to show that there are fundamental limitations on the kinds of patterns that perceptrons can ever learn to recognize."



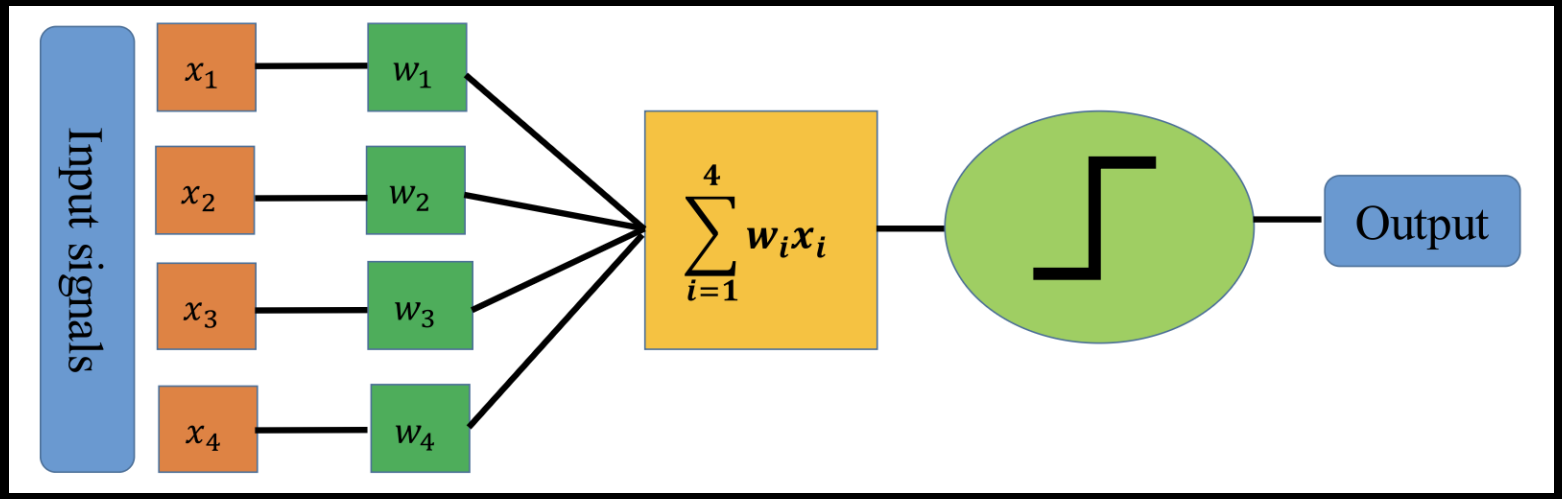
Motivation: Act of Service

"Rosenblatt's (1958a) schemes quickly took root, and soon there were perhaps as many as a hundred groups, large and small, experimenting with the model... The results… were generally disappointing… Both of the present authors (first independently and later together) became involved with a somewhat therapeutic compulsion: to dispel what we feared to be the first shadows of a 'holistic' or 'Gestalt' misconception that would threaten to haunt the fields of engineering and artificial intelligence as it had earlier haunted biology and psychology."
– pg. 18-20



Context

- Working generalizable perceptrons at the time had a single layer; e.g.,



[Ghasemian, Razminia, and Rostami. Quantum Information Processing 2023]

- Their analysis excluded incomplete visions from Rosenblatt and others for “multilayer perceptrons” (e.g., see [Rosenblatt. Spartan books 1962])

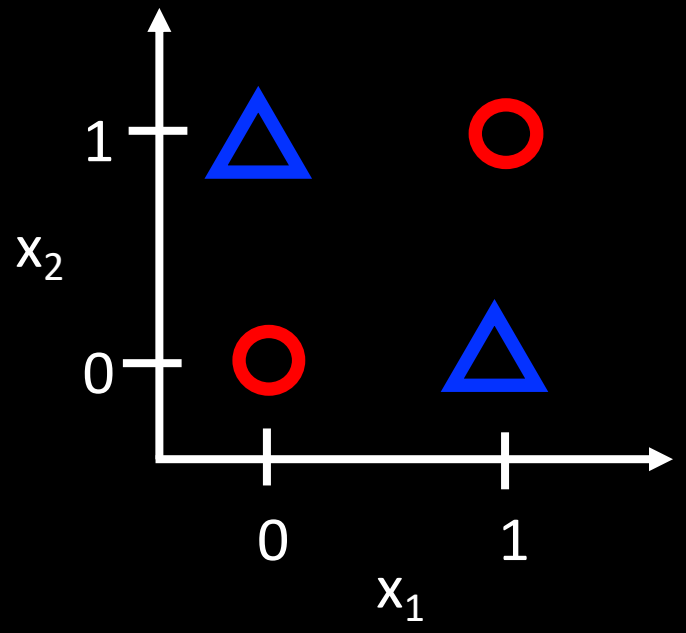
Example: XOR for Single Layer Perceptron

XOR = “Exclusive Or”

- Input: two binary values x_1 and x_2
- Output:
  - 1, when exactly one input equals 1
  - 0, otherwise

x_1	x_2	x_1 XOR x_2
0	0	?
0	1	?
1	0	?
1	1	?

Example: XOR Problem Visualized Graphically



x_1	x_2	x_1 XOR x_2
0	0	0
0	1	1
1	0	1
1	1	0

Any volunteers for drawing a line (decision boundary from a perceptron for 2 features) to separate the two classes?

Many more problems were also shown impossible to solve with a single layer perceptron

[Minsky and Papert. 1969]

~~X is not a single point.~~) One way to prove this is to deduce a contradiction from the hypothesis that numbers α, β, and θ can be found for which

$$\psi_{\equiv}(x,y) = \text{~~}xy \vee \bar{x}\bar{y}\text{~~} = [\alpha x + \beta y > \theta].$$

We can proceed directly by writing down the conditions on α and β:

$$\psi_{\equiv}(1,0) = 0 \implies \alpha \leq \theta \qquad \text{(because } \alpha \cdot 1 + \beta \cdot 0 \leq \theta\text{)}$$
$$\psi_{\equiv}(0,1) = 0 \implies \beta \leq \theta$$
$$\psi_{\equiv}(1,1) = 1 \implies \alpha + \beta > \theta$$
$$\psi_{\equiv}(0,0) = 1 \implies 0 > \theta$$

In this simple case it is easy enough to deduce the contradiction, for adding the first two conditions gives us

$$\alpha + \beta \leq 2\theta,$$

and this, with the third implies that

$$\theta < 2\theta,$$

and this would make θ positive, contradicting the fourth condition.

Helped Start an “AI Winter” (1970s to mid-1980s)

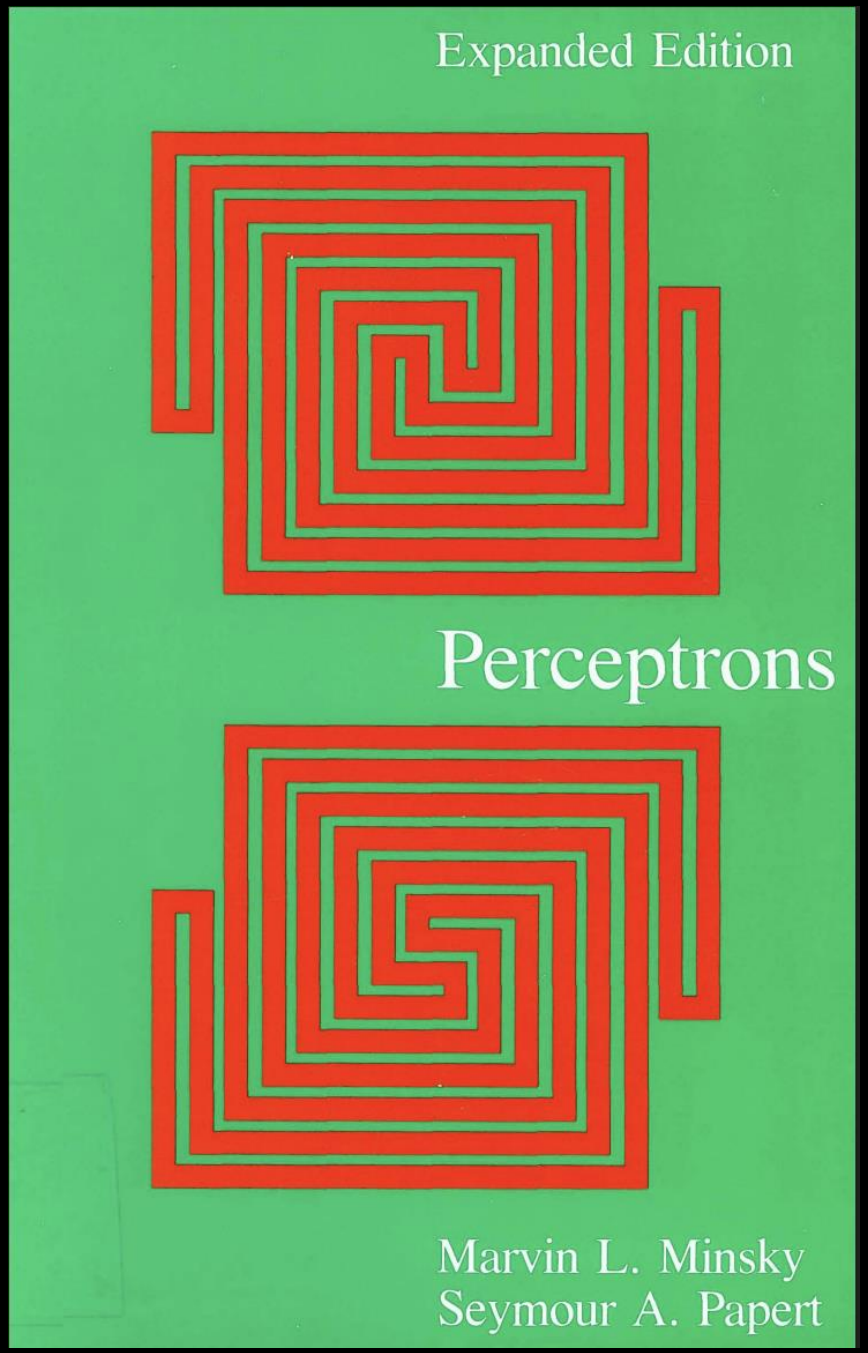


- Funding agencies lost interest
- Researchers lost interest and left for other topics

https://medium.com/@bergholz/proclaimed-dead-twice-the-ai-winters-5848c7f8a29f

Retrospective Thoughts

"How did the scientists involved with such matters react to this? One popular version is that the publication of our book so discouraged research on learning in network machines that a promising line of research was interrupted. Our version is that progress had already come to a virtual halt because of the lack of adequate basic theories, and the lessons in this book provided the field with new momentum-albeit, paradoxically, by redirecting its immediate concerns. To understand the situation, one must recall that by the mid 1960s there had been a great many experiments with perceptrons, but no one had been able to explain why they were able to learn to recognize certain kinds of patterns and not others. Was this in the nature of the learning procedures? Did it depend on the sequences in which the patterns were presented? Were the machines simply too small in capacity?"



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Context: Preparation for Lab Assignments

- Lab assignment 1 manual (released at 1:45pm on Canvas): programming skills needed will be covered in the programming tutorials
- Assigned readings: covers basic ideas addressed in the lab assignments, including Thursday's assigned reading for assignment 1

Today’s Topics

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The End